

2.7-2.8 补充 (1)

Lemma (算术基本定理的推论). n 是一个任意的非零整数, 则 n 有素因子分解:

$$n = \pm p_1^{a_1} \cdots p_r^{a_r}$$

其中 $r \in \mathbb{Z}_{\geq 0}$ (当 $r=0$ 时右式规定为 ± 1), $p_1, \dots, p_r$ 是相异素数, $a_1, \dots, a_r \in \mathbb{Z}_{\geq 1}$,

而且此分解不论顺序是唯一的. d 是一个任意的非零整数, 则有:

$$d \mid n \iff d = \pm p_1^{e_1} \cdots p_r^{e_r}, \text{ 其中 } e_1, \dots, e_r \in \mathbb{Z}_{\geq 0}, e_i \leq a_1, \dots, e_r \leq a_r$$

($a_1, \dots, a_r \in \mathbb{Z}_{\geq 0}$ 时本引理也成立)

Proof: ($\Leftarrow$): $\because d = \pm p_1^{e_1} \cdots p_r^{e_r}$, 其中 $e_1, \dots, e_r \in \mathbb{Z}_{\geq 0}, e_i \leq a_1, \dots, e_r \leq a_r$

$$\therefore d \in \mathbb{Z} \text{ 且 } d \neq 0.$$

$$\therefore \frac{n}{d} = \frac{\pm p_1^{a_1} \cdots p_r^{a_r}}{\pm p_1^{e_1} \cdots p_r^{e_r}} = \pm \frac{p_1^{a_1}}{p_1^{e_1}} \cdot \frac{p_2^{a_2}}{p_2^{e_2}} \cdots \frac{p_r^{a_r}}{p_r^{e_r}} = \pm p_1^{a_1-e_1} p_2^{a_2-e_2} \cdots p_r^{a_r-e_r}$$

由 n 和 d 的正负来确定

其中 $a_1 - e_1 \in \mathbb{Z}_{\geq 0}, a_2 - e_2 \in \mathbb{Z}_{\geq 0}, \dots, a_r - e_r \in \mathbb{Z}_{\geq 0}$

$\therefore \frac{n}{d}$ 是非零整数. $\therefore d \mid n$

($\Rightarrow$): $\because d$ 是一个任意的非零整数 $\therefore$ 对 d 分类讨论如下:

~~① 当 $d=1$ 时, 此时 $d \mid n$ 成立. 此时对应于 $r=0$ 的情形. 结论得证~~

~~② 当 $d=-1$ 时, 此时 $d \mid n$ 成立. 此时~~

① 当 $d=1$ 时, 此时 $d \mid n$ 成立. 此时 $d = +p_1^0 \cdots p_r^0$ (即 $e_1 = \dots = e_r = 0$)

② 当 $d=-1$ 时, 此时 $d \mid n$ 成立. 此时 $d = -p_1^0 \cdots p_r^0$ (即 $e_1 = \dots = e_r = 0$)

③ 当 $d \geq 2$ 时, 设 λ 是 d 的任意一个素因子. 则有: λ 是素数且 $\lambda \mid d$.

$$\therefore \lambda \mid d \text{ 且 } d \mid n \quad \therefore \lambda \mid n \quad \therefore \lambda \mid \pm p_1^{a_1} \cdots p_r^{a_r}$$

$$\therefore \lambda \mid \underbrace{p_1 \cdots p_1}_{a_1 \uparrow} \underbrace{p_2 \cdots p_2}_{a_2 \uparrow} \cdots \underbrace{p_r \cdots p_r}_{a_r \uparrow} \quad \therefore \lambda \mid p_1 \text{ 或 } \lambda \mid p_2 \text{ 或 } \cdots \text{ 或 } \lambda \mid p_r$$

$$\therefore \lambda = p_1 \text{ 或 } \lambda = p_2 \text{ 或 } \dots \text{ 或 } \lambda = p_r$$

$$\therefore d = \pm p_1^{e_1} p_2^{e_2} \dots p_r^{e_r}, \text{ 其中 } e_1, e_2, \dots, e_r \in \mathbb{Z}_{\geq 0}$$

$$\text{假设 } e_1 > a_1. \text{ 则有: } \therefore d | n \quad \therefore \frac{n}{d} \in \mathbb{Z}$$

$$\therefore \frac{n}{d} = \frac{\pm p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}}{\pm p_1^{e_1} p_2^{e_2} \dots p_r^{e_r}} = \pm p_1^{a_1 - e_1} p_2^{a_2 - e_2} \dots p_r^{a_r - e_r} = \pm p_1^{-(e_1 - a_1)} p_2^{a_2 - e_2} \dots p_r^{a_r - e_r}$$

$$\equiv \pm \frac{p_2^{a_2 - e_2} \dots p_r^{a_r - e_r}}{p_1^{e_1 - a_1}} \in \mathbb{Z} \quad \text{其中 } e_1 - a_1 \in \mathbb{Z}_{\geq 1}$$

$$\therefore \frac{n}{d} = \frac{\pm p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}}{\pm p_1^{e_1} p_2^{e_2} \dots p_r^{e_r}} = \pm \frac{p_2^{a_2} \dots p_r^{a_r}}{p_1^{e_1 - a_1} p_2^{e_2} \dots p_r^{e_r}} \in \mathbb{Z}$$

$$\therefore p_1^{e_1 - a_1} p_2^{e_2} \dots p_r^{e_r} \mid p_2^{a_2} \dots p_r^{a_r}, \text{ 其中 } e_1 - a_1 \in \mathbb{Z}_{\geq 1}$$

$$\therefore p_1 \mid p_1^{e_1 - a_1} p_2^{e_2} \dots p_r^{e_r}, \quad p_1^{e_1 - a_1} p_2^{e_2} \dots p_r^{e_r} \mid p_2^{a_2} \dots p_r^{a_r} \quad \therefore p_1 \mid p_2^{a_2} \dots p_r^{a_r}$$

$$\therefore p_1 \mid p_2 \text{ 或 } p_1 \mid p_3 \text{ 或 } \dots \text{ 或 } p_1 \mid p_r \quad \therefore p_1 = p_2 \text{ 或 } p_1 = p_3 \text{ 或 } \dots \text{ 或 } p_1 = p_r. \text{ 矛盾. } \therefore e_1 \leq a_1$$

$$\text{假设 } e_2 > a_2. \text{ 则有: } \therefore d | n \quad \therefore \frac{n}{d} \in \mathbb{Z}$$

$$\therefore \frac{n}{d} = \frac{\pm p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots p_r^{a_r}}{\pm p_1^{e_1} p_2^{e_2} p_3^{e_3} \dots p_r^{e_r}} = \pm \frac{p_1^{a_1} p_3^{a_3} \dots p_r^{a_r}}{p_1^{e_1} p_2^{e_2 - a_2} p_3^{e_3} \dots p_r^{e_r}} \in \mathbb{Z}$$

$$\therefore p_1^{e_1} p_2^{e_2 - a_2} p_3^{e_3} \dots p_r^{e_r} \mid p_1^{a_1} p_3^{a_3} \dots p_r^{a_r}, \text{ 其中 } e_2 - a_2 \in \mathbb{Z}_{\geq 1}$$

$$\therefore p_2 \mid p_1^{e_1} p_2^{e_2 - a_2} p_3^{e_3} \dots p_r^{e_r}, \quad p_1^{e_1} p_2^{e_2 - a_2} p_3^{e_3} \dots p_r^{e_r} \mid p_1^{a_1} p_3^{a_3} \dots p_r^{a_r} \quad \therefore p_2 \mid p_1^{a_1} p_3^{a_3} \dots p_r^{a_r}$$

$$\therefore p_2 \mid p_1 \text{ 或 } p_2 \mid p_3 \text{ 或 } \dots \text{ 或 } p_2 \mid p_r$$

$$\therefore p_2 = p_1 \text{ 或 } p_2 = p_3 \text{ 或 } \dots \text{ 或 } p_2 = p_r. \text{ 矛盾. } \therefore e_2 \leq a_2$$

完全类似地, 可以证明 $e_3 \leq a_3, \dots, e_r \leq a_r$

$\therefore d = \pm p_1^{e_1} p_2^{e_2} \dots p_r^{e_r}$, 其中 $e_1, e_2, \dots, e_r \in \mathbb{Z}_{\geq 0}$, $e_1 \leq a_1, e_2 \leq a_2, \dots, e_r \leq a_r$.

□

Lemma (素因子分解后求两个数的最大公因数) 设 $n = \pm \prod_{i=1}^r p_i^{a_i}$ 和 $m = \pm \prod_{i=1}^r p_i^{b_i}$ 是两个整数,

其中 $r \in \mathbb{Z}_{\geq 0}$, $p_1, \dots, p_r$ 是相异素数, $a_i, b_i \in \mathbb{Z}_{\geq 0}$, 则有:

$$\gcd(n, m) = \prod_{i=1}^r p_i^{\min\{a_i, b_i\}}$$

Proof: 当 $r=0$ 时, $n=\pm 1, m=\pm 1$, $\gcd(n, m) = 1 = \prod_{i=1}^0 p_i^{\min\{a_i, b_i\}}$, 结论得证.

当 $r \in \mathbb{Z}_{\geq 1}$ 时, n 是非零整数, m 是非零整数

$$\therefore \gcd(n, m) = \max\{d \in \mathbb{Z}_{\geq 1} : d|n \text{ 且 } d|m\} \in \mathbb{Z}_{\geq 1}.$$

对任意的非零整数 d , 有:

$$d|n \text{ 且 } d|m$$

$$\Leftrightarrow \left(d = \pm \prod_{i=1}^r p_i^{e_i}, \text{ 其中 } e_i \in \mathbb{Z}_{\geq 0}, e_i \leq a_i \right) \text{ 且 } \left(d = \pm \prod_{i=1}^r p_i^{e_i}, \text{ 其中 } e_i \in \mathbb{Z}_{\geq 0}, e_i \leq b_i \right)$$

$$\Leftrightarrow d = \pm \prod_{i=1}^r p_i^{e_i}, \text{ 其中 } e_i \in \mathbb{Z}_{\geq 0}, e_i \leq \min\{a_i, b_i\}.$$

$$\therefore \gcd(n, m) = \max\{d \in \mathbb{Z}_{\geq 1} : d|n \text{ 且 } d|m\}$$

$$= \max\{d \in \mathbb{Z}_{\geq 1} : d = \prod_{i=1}^r p_i^{e_i}, \text{ 其中 } e_i \in \mathbb{Z}_{\geq 0}, e_i \leq \min\{a_i, b_i\}\}$$

$$= \prod_{i=1}^r p_i^{\min\{a_i, b_i\}}$$

($\because$ 对 $\forall i=1, \dots, r$, $p_i \geq 2$ ~~且~~ 函数 $y = p_i^x$ 在 $\mathbb{R}$ 上严格单调增.

$$\therefore p_i^{e_i} \leq p_i^{\min\{a_i, b_i\}} \quad \therefore 1 \leq p_i^{e_i} \leq p_i^{\min\{a_i, b_i\}}$$

$$\therefore 1 \leq \prod_{i=1}^r p_i^{e_i} \leq \prod_{i=1}^r p_i^{\min\{a_i, b_i\}}$$

$$\therefore \max \{ d \in \mathbb{Z}_{\geq 1} : d = \prod_{i=1}^r p_i^{e_i}, \text{ 其中 } e_i \in \mathbb{Z}_{\geq 0}, e_i \leq \min\{a_i, b_i\} \} = \prod_{i=1}^r p_i^{\min\{a_i, b_i\}}$$



~~Lemma (任意非零整数的倍数的充要条件) $n = \pm \prod_{i=1}^r p_i^{a_i}$ 是一个整数, 其中 $r \in \mathbb{Z}_{\geq 0}$,~~

~~$p_1, \dots, p_r$ 是相异素数, $a_i \in \mathbb{Z}_{\geq 0}$. d 是一个任意的非零整数, 则有:~~

~~$$n|d \Leftrightarrow d = \pm \prod_{i=1}^k p_i^{e_i}, \text{ 其中 } k \in \mathbb{Z}_{\geq 0}, k \geq r, e_i \in \mathbb{Z}_{\geq 0}, e_1 \geq a_1, \dots, e_r \geq a_r.$$~~

~~Proof: ($\Leftarrow$): $\therefore d = \pm \prod_{i=1}^k p_i^{e_i}$, 其中 $k \in \mathbb{Z}_{\geq 0}, k \geq r, e_i \in \mathbb{Z}_{\geq 0}, e_1 \geq a_1, \dots, e_r \geq a_r.$~~

~~$$\therefore d \in \mathbb{Z} \text{ 且 } d \neq 0.$$~~

~~$$\therefore \frac{d}{n} = \frac{\pm p_1^{e_1} \dots p_r^{e_r} \dots p_k^{e_k}}{\pm p_1^{a_1} \dots p_r^{a_r}} = \pm p_1^{e_1 - a_1} \dots p_r^{e_r - a_r} \dots p_k^{e_k} \text{ 是非零整数.}$$~~

~~$$\therefore n|d$$~~

~~$$(\Rightarrow): \therefore d \text{ 是一个任意的非零整数, } n|d$$~~

~~$$\therefore \exists \varphi \in \mathbb{Z}, \varphi \neq 0, \text{ s.t. } d = n\varphi,$$~~

~~$$\therefore d = \left(\pm \prod_{i=1}^r p_i^{a_i} \right) \cdot \varphi.$$~~

~~将 φ 分解成素因数的乘积. 若某个素因子等于某个 p_i (其中 $i=1, \dots, r$).~~

~~则 d 的素因子分解中包含 $p_i^{e_i}$ (其中 $e_i \geq a_i$) 一项.~~

~~若 φ 的某个素因子不等于 $p_1, \dots, p_r$, 则将这个素因子记作 $p_{r+1}, p_{r+2}, \dots$ 即可.~~



Lemma (任意非零整数的倍数的充要条件) $n = \pm \prod_{i=1}^r p_i^{a_i}$ 是一个整数, 其中 $r \in \mathbb{Z}_{\geq 0}$,

$p_1, \dots, p_r$ 是相异素数, $a_1, \dots, a_r \in \mathbb{Z}_{\geq 0}$. d 是一个任意的非零整数, 则有:

$$n|d \Leftrightarrow d = \pm \prod_{i=1}^r p_i^{e_i} \prod_{j=1}^s q_j^{\alpha_j}, \text{ 其中 } p_1, \dots, p_r, q_1, \dots, q_s \text{ 是相异素数,}$$

$$s \in \mathbb{Z}_{\geq 0}, e_i, \alpha_j \in \mathbb{Z}_{\geq 0}, e_i \geq a_i.$$

Proof: ($\Leftarrow$): $\because d = \pm \prod_{i=1}^r p_i^{e_i} \prod_{j=1}^s q_j^{\alpha_j}$, 其中 $p_1, \dots, p_r, q_1, \dots, q_s$ 是相异素数,

$$s \in \mathbb{Z}_{\geq 0}, e_i, \alpha_j \in \mathbb{Z}_{\geq 0}, e_i \geq a_i$$

$$\therefore d \in \mathbb{Z} \text{ 且 } d \neq 0$$

$$\because n = \pm \prod_{i=1}^r p_i^{a_i} \quad \therefore n \in \mathbb{Z} \text{ 且 } n \neq 0$$

$$\therefore \frac{d}{n} = \frac{\pm \prod_{i=1}^r p_i^{e_i} \prod_{j=1}^s q_j^{\alpha_j}}{\pm \prod_{i=1}^r p_i^{a_i}} = \pm \prod_{i=1}^r p_i^{e_i - a_i} \prod_{j=1}^s q_j^{\alpha_j}, \text{ 其中 } e_i - a_i \in \mathbb{Z}_{\geq 0}$$

$$\therefore \frac{d}{n} \in \mathbb{Z} \text{ 且 } \frac{d}{n} \neq 0. \quad \therefore n|d.$$

($\Rightarrow$): $\because d$ 是一个任意的非零整数, $n|d$

$$\therefore \exists \theta \in \mathbb{Z}, \text{ s.t. } d = n\theta \quad \because d \neq 0 \quad \therefore \theta \in \mathbb{Z} \text{ 且 } \theta \neq 0$$

$$\therefore \theta \text{ 有素因子分解: } \theta = \pm \prod_{i=1}^r p_i^{\beta_i} \prod_{j=1}^s q_j^{\alpha_j}, \quad \beta_1, \dots, \beta_r \in \mathbb{Z}_{\geq 0}, \alpha_1, \dots, \alpha_s \in \mathbb{Z}_{\geq 0}.$$

$p_1, \dots, p_r, q_1, \dots, q_s$ 是相异素数, $s \in \mathbb{Z}_{\geq 0}$.

(即: θ 的素因子分与 n 的素因子相同的和与 n 的素因子不同的来考虑)

$$\therefore d = n\theta = \pm \prod_{i=1}^r p_i^{a_i} \prod_{i=1}^r p_i^{\beta_i} \prod_{j=1}^s q_j^{\alpha_j} = \pm \prod_{i=1}^r p_i^{e_i} \prod_{j=1}^s q_j^{\alpha_j}.$$

其中 $p_1, \dots, p_r, q_1, \dots, q_s$ 是相异素数, $s \in \mathbb{Z}_{\geq 0}, e_i, \alpha_j \in \mathbb{Z}_{\geq 0}, e_i \geq a_i$



Lemma (素因子分解后求两个数的最小公倍数) 设 $n = \pm \prod_{i=1}^r p_i^{a_i}$ 和 $m = \pm \prod_{i=1}^r p_i^{b_i}$ 是两个整数, 其中 $r \in \mathbb{Z}_{\geq 0}$, $p_1, \dots, p_r$ 是相异素数, $a_i, b_i \in \mathbb{Z}_{\geq 0}$. 则有:

$$\text{lcm}(n, m) = \prod_{i=1}^r p_i^{\max\{a_i, b_i\}}$$

Proof: 当 $r=0$ 时, $n = \pm 1$, $m = \pm 1$, $\text{lcm}(n, m) = 1 = \prod_{i=1}^0 p_i^{\max\{a_i, b_i\}}$, 结论得证.

当 $r \in \mathbb{Z}_{\geq 1}$ 时, $\therefore n \in \mathbb{Z}$ 且 $n \neq 0$, $m \in \mathbb{Z}$ 且 $m \neq 0$

$$\therefore \text{lcm}(n, m) = \min\{d \in \mathbb{Z}_{\geq 1} : n|d \text{ 且 } m|d\}.$$

对 $\forall d \in \mathbb{Z}$ 且 $d \neq 0$, 有:

$$n|d \text{ 且 } m|d$$

$$\Leftrightarrow d = \pm \prod_{i=1}^r p_i^{e_i} \prod_{j=1}^s q_j^{\alpha_j}, \text{ 其中 } p_1, \dots, p_r, q_1, \dots, q_s \text{ 是相异素数, } s \in \mathbb{Z}_{\geq 0}, e_i, \alpha_j \in \mathbb{Z}_{\geq 0}, e_i \geq a_i$$

$$\text{且 } d = \pm \prod_{i=1}^r p_i^{h_i} \prod_{j=1}^t u_j^{\beta_j}, \text{ 其中 } p_1, \dots, p_r, u_1, \dots, u_t \text{ 是相异素数, } t \in \mathbb{Z}_{\geq 0}, h_i, \beta_j \in \mathbb{Z}_{\geq 0}, h_i \geq b_i$$

$$\Leftrightarrow d = \pm \prod_{i=1}^r p_i^{\gamma_i} \prod_{j=1}^w v_j^{\varepsilon_j}, \text{ 其中 } p_1, \dots, p_r, v_1, \dots, v_w \text{ 是相异素数, } w \in \mathbb{Z}_{\geq 0}, \gamma_i, \varepsilon_j \in \mathbb{Z}_{\geq 0},$$

$$\gamma_i \geq \max\{a_i, b_i\}.$$

$$\therefore \text{lcm}(n, m) = \min\{d \in \mathbb{Z}_{\geq 1} : n|d \text{ 且 } m|d\}$$

$$= \min\{d \in \mathbb{Z}_{\geq 1} : d = \prod_{i=1}^r p_i^{\gamma_i} \prod_{j=1}^w v_j^{\varepsilon_j}, \text{ 其中 } p_1, \dots, p_r, v_1, \dots, v_w \text{ 是相异素数, } w \in \mathbb{Z}_{\geq 0}, \gamma_i, \varepsilon_j \in \mathbb{Z}_{\geq 0},$$

$$\gamma_i \geq \max\{a_i, b_i\}\}$$

$$= \prod_{i=1}^r p_i^{\max\{a_i, b_i\}} \prod_{j=1}^0 v_j^{\varepsilon_j} = \prod_{i=1}^r p_i^{\max\{a_i, b_i\}}$$

